Reading Guide to the DROS Research Trajectory: An Agent Runtime Operation Substrate across Digital and Physical Domains

Chun-Cheng (Jimmy) Chen
Top-Celestial Company Ltd.
Taipei, Taiwan R.O.C.
jimmychen@dr-os.io

Abstract—This Technical Note provides an overarching reading guide to the DROS Research Trajectory, systematically expanding a single foundational scientific question across five progressive scales of execution governance. From high-level enterprise trust boundaries and forensic auditability down to binary in-band interception, mobile operating system API isolation, and cyber-physical drone kinetic envelope containment, the series is anchored by the open-source DROS-VEP Lite research vessel. This guide clarifies the scientific scope, reading paths, and authoritative DOI citation identifiers across all five core technical papers.

Index Terms—Agent Runtime Governance, Post-Compromise Security, Execution Authority, Physical AI, Mobile Security, Evaluation Protocol.

I. Purpose and The 5 Research Scales

The DROS research trajectory expands a singular, foundational scientific question across five progressive scales of execution governance, supported by an open, reproducible evaluation vessel (DROS-VEP Lite):

$$\boxed{\text{Governance} \rightarrow \text{Authority} \rightarrow \text{Runtime} \rightarrow \text{Digital} \rightarrow \text{Physical}} \tag{1}$$

These five scales construct an end-to-end causal defense chain:

1) Governance Model (6P): Enterprise trust delegation and revocation boundaries.
2) Runtime Architecture (4-Layer): Deterministic policy execution and forensic Merkle auditability.
3) Kernel Control (PGM): Post-compromise binary C-ABI microsecond hard tripping.
4) Digital Substrate (Mobile): Privilege isolation across iOS/Android OS and protected APIs.
5) Physical Substrate (UAV): Kinetic envelope containment and lookahead braking in robotics.

II. The Overarching Research Thesis

$$\boxed{\begin{array}{c}\text{Compromise(Cognitive Controller)} \\ \not\Rightarrow \text{Compromise(Downstream Execution Authority)}\end{array}} \tag{2}$$

> "A compromise of the autonomous cognitive controller does not inherently entail a compromise of downstream execution authority."

Rather than relying exclusively on internal statistical alignment of generative models, the DROS research trajectory investigates the formal decoupling of cognitive intent from actual execution, establishing verifiable deterministic execution gates within declared enforcement boundaries that cannot be arbitrarily subverted by untrusted cognitive planes.

III. Substrate vs. Protocol Positioning (The VEP Vessel)

This series formalizes the research problem as an executable, attackable, and measurable experimental substrate problem:

1) VEP Research Vessel (Evaluation Protocol): Formalizes vendor-neutral RFC-001 Execution Governance Evaluation Standards, providing multi-architecture baselines, 72-hour soak tests, and an open falsification channel.
2) DROS Runtime (Reference Substrate): Operates as a concrete, deterministic binary reference implementation under the VEP specification.
3) Dual-Domain Instantiations (Digital & Physical):
   - Mobile: Evaluates Unauthorized Protected System-Effect Invariance ($\text{Auth} = 0 \implies \text{PSE} \equiv 0$) and microsecond revocation races ($T_{\text{rev}} < 2.5\ \mu\text{s}$).
   - UAV: Evaluates Unauthorized Command Invariance ($\text{Auth} = 0 \implies \Delta u_{\text{cmd}} \equiv 0$) and physical safety-envelope preservation ($\mathcal{S}_{\text{safe}}$).

IV. Cross-Domain Substrate Mapping

DROS demonstrates that deterministic runtime governance applies uniformly across heterogeneous operational substrates:

- Digital Domains (Mobile OS): Protects privileged APIs, clipboard, storage, and sensors against unauthorized state leakage.
- Physical Domains (Robotics & UAVs): Enforces physical kinetic invariance, dynamic deceleration horizons, and geofence boundary containment.

TABLE I
The DROS Research Trajectory: Core Technical Papers and Evaluation Vessel Matrix

Component	Paper / System Title	Role in Research Trajectory	Core Question / Functional Scope	Reference / DOI
Vessel	DROS-VEP Lite	Evaluation Protocol & Sandbox	How to provide an objective, falsifiable evaluation testbed?	GitHub Official Repo
Paper 1	DROS-6P	Governance Model	Who has authority? Why? When does it expire? (Enterprise Trust)	10.5281/zenodo.21833970
Paper 2	DROS 4-Layer	Runtime Architecture	How does policy deterministically reach execution? (Forensic Merkle)	10.5281/zenodo.22092008
Paper 3	DROS-PGM	Kernel-Level Control	Once compromised, can the execution control plane still block action?	10.5281/zenodo.21903687
Paper 4	Post-Compromise Mobile	Digital Substrate	Does execution authority hold across Mobile OS/APIs?	10.5281/zenodo.22253147
Paper 5	Post-Compromise UAV	Physical Substrate	When actions cause physical motion, can the kinetic envelope hold?	10.5281/zenodo.22254372

V. Recommended Reading Paths

- Architects & Decision Makers: Start with this Guide → explore the DROS-VEP Lite Root README.
- Governance & Policy Researchers: DROS-6P → DROS 4-Layer.
- Security Engineers & Binary Specialists: DROS-PGM → Post-Compromise Mobile.
- Robotics & Control Theorists: Post-Compromise UAV (Kinematics, braking horizon, latency decoupling).

VI. References & Identifiers

When referencing technical claims, cite via the official DOI or technical report identifier:

1) DROS-VEP Lite: Verifiable Execution Protocol & Evaluation Sandbox, 2026. https://github.com/Top-Celestial-Company-Ltd/DROS-VEP-lite
2) DROS-6P: DOI: 10.5281/zenodo.21833970
3) DROS 4-Layer: DOI: 10.5281/zenodo.22092008
4) DROS-PGM: DOI: 10.5281/zenodo.21903687
5) DROS-Mobile: DOI: 10.5281/zenodo.22253147
6) DROS-Kinetic (UAV): DOI: 10.5281/zenodo.22254372